MILITARY SPECIFICATION

TROUSERS, PNEUMATIC, ANTI-SHOCK, Universal Size, Blue

This specification is approved for use by the Defense Personnel Support Center, Defense Logistics Agency, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers anti-shock pneumatic trousers, with two inflatable chambers for applying pneumatic pressure to the abdomen and legs, for use in emergency first-aid treatment to relieve shock, restore blood pressure and counteract internal bleeding.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636	- Boxes, Shipping, Fiberboard.
QQ-W-423	- Wire, Steel, Corrosion-Resisting.
V-T-295	- Thread, Nylon.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, CODE-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6515

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:24Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-A-8625	- Anodic Coatings, for Aluminum and aluminum alloys.
MIL-C-508	- Cloth, Oxford, Nylon, 3 ounces.
MIL-C-4150	- Case, Transit and Storage, Waterproof and Water-vaporproof.
MIL-F-21840	- Fastener Tapes, Hook and Pile, Synthetic.
MIL-R-3065	- Rubber, fabricated parts.
MIL-R-6855	- Rubber, Synthetic Sheets, Strips, Molded or Extruded shapes.

STANDARD

FEDERAL

FED-STD-191	- Textile Test Method.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurements should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. Materials used shall be of the highest quality and shall be suitable in all respects for the purpose intended.

3.1.1 Garment. The garment shall be fabricated from commercial grade polyurethane coated, two-ply taffeta heat sealable nylon cloth.

3.1.2 Rubber tubing. The rubber tubing shall be in accordance with MIL-R-6855, class II, grade 60.

3.1.3 Acetate resin. The acetate resin shall be type TSO-C13, and shall be in accordance with MIL-C-4150.

3.1.4 Pressure relief valves. The pressure relief valve body shall be fabricated from anodized aluminum alloy 2011-T3, as specified in MIL-A-8625, for type II, clear. The valve screen shall be fabricated from type 302 or 304 corrosion-resistant steel, as specified in QQ-W-423.

3.1.5 Manually adjusted valves. The manually adjusted valves shall be fabricated from commercial molded polypropylene.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:24Z
Check the source to verify that this is the current version before use.

3.1.6 Inflation tubing. The inflation tubing shall be fabricated of neoprene, as specified in MIL-R-3065.

3.1.7 "Y" connectors. The "Y" connectors shall be fabricated of commercial grade polyethylene.

3.1.8 Foot operated pump. The dome shaped cavity of the foot operated pump shall be fabricated of Buna-N rubber, AMS-3210, in accordance with MIL-R-6855.

3.1.9 Air inlet valve. The air inlet valve housing shall be fabricated of commercial molded nylon.

3.1.10 Air exit valve. The air exit valve housing shall be fabricated from 2011-T3 anodized aluminum alloy in accordance with MIL-A-8625, type II, clear.

3.1.11 Carrying case. The carrying case shall be fabricated of commercial grade ABS plastic.

3.2 Design and construction. The pneumatic trousers shall consist of the garment, inflation tubing, three manually adjusted valves, two "Y" connectors, two pressure relief valves, a foot operated air pump, operating instructions and a carrying case. The tubing, "Y" connectors and pneumatic fittings shall have a minimum 1/4 inch diameter.

3.2.1 Garment. The garment shall be heat sealed in the pattern as shown in figure 1. The hook and pile fastener pads shall be in accordance with MIL-F-21840, type II, class 1, and MIL-F-21840, class 1 respectively. The fastener pads and binding shall be sewn to the garment with nylon thread in accordance with MIL-C-508. At the locations shown in figure 1, the nylon oxford textile tape binding shall be in accordance with V-T-295, class 1, type I. The garment shall have two air chambers, one for both legs that shall be joined pneumatically, and one for the abdomen. Both chambers shall be individiually controlled. The garment material shall be blue in color.

3.2.2 Pneumatic system components. The pneumatic system components shall be interconnected to each other and the garment as shown in figure 1. Rubber tubing shall be attached to the garment with acetate resin. The rubber tubing sections shall be secured to the component parts by plastic ties or other suitable means.

3.2.2.1 Foot operated pump. The foot operated pump shall consist of a closed dome shaped cavity, an air inlet valve and an air exit valve. The air inlet valve shall be provided with a check valve to prevent air from being expelled through the inlet valve in the pump pressure stroke. The air exit valve shall be provided with a check valve to prevent air from the pressurized garment from reentering the pump. The dome shaped cavity shall be 6-1/4 $\pm$ 1/4 inches in diameter and 3-1/4 $\pm$ 1/4 inches high.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:24Z
Check the source to verify that this is the current version before use.

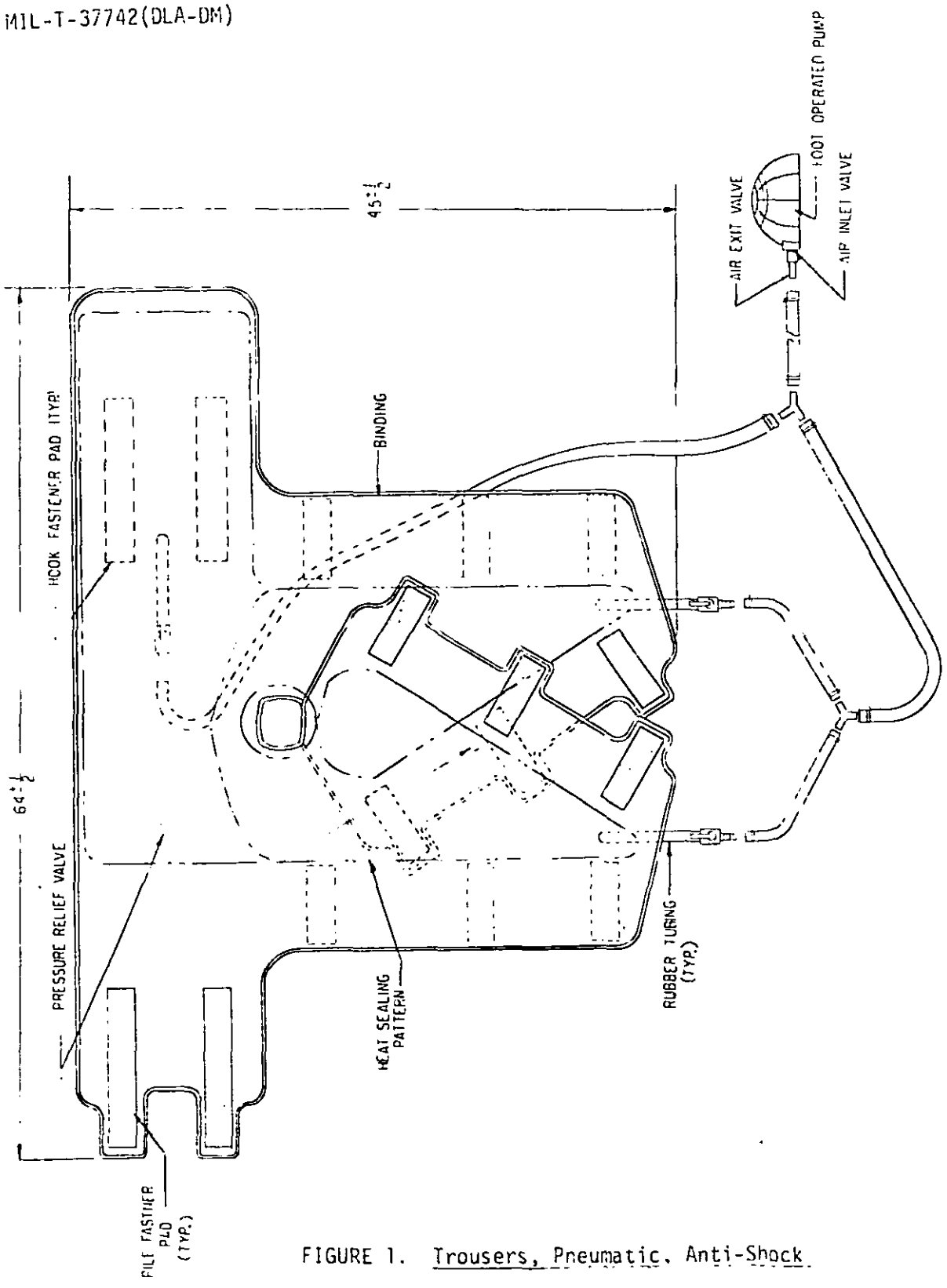


FIGURE 1. Trousers, Pneumatic, Anti-Shock

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:24Z
Check the source to verify that this is the current version before use.

3.3 Carrying case. The carrying case dimensions shall be 13-1/2 ± 1/4 inches long, 10 ± 1/4 inches wide by 5-1/4 ± 1/4 inches high. The pneumatic trousers and all components shown in figure 1 shall fit in the closed carrying case. The carrying case shall be provided with a handle and latching device.

3.4 Weight. The overall weight of the garment material shall be not less than 4.70 ounce per square yard when tested as specified (see 4.4.1).

3.5 Breaking strength.

3.5.1 Warp. The breaking strength of the warp shall be not less than 100 pounds when tested as specified (see 4.4.2).

3.5.2 Filling. The breaking strength of the filling shall be not less than 90 pounds when tested as specified (see 4.4.3).

3.6 Leakage. The cloth shall show no signs of air leakage when tested as specified (see 4.4.4).

3.7 Inflation. The number of foot pump strokes necessary to inflate both chambers of the garment to 104 mm of Hg, shall not exceed 36 strokes when tested as specified (see 4.4.5).

3.8 Pressure tests.

3.8.1 Relief valve operation. Each pressure relief valve shall operate at a pressure of 120 mm ± 5 mm of Hg, when tested as specified (see 4.4.6).

3.8.2 Sustained pressure. The trousers shall be capable of maintaining an air pressure of at least 80 mm of Hg, for at least 30 minutes when tested as specified (see 4.4.7).

3.8.3 Proof pressure. The garment shall be capable of sustaining a pressure of 130 mm of Hg, with no damage or leakage for a period of 15 seconds when tested as specified (see 4.4.8).

3.9 Instructions. Instructions for use of the pneumatic trousers shall be attached to the inside top surface of the carrying case cover.

3.10 Identification marking. The cover of each carrying case shall be marked with the name of the item and the name or registered trademark of the manufacturer.

3.11 Workmanship. Workmanship shall be first class throughout. All components shall be free from defects which detract from their appearance or may impair their serviceability.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:24Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificate of quality. Certificate of quality, supplied by the manufacturer of the pneumatic trouser material may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I.

TABLE I. Sampling for examination.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-2	2.5

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:24Z
Check the source to verify that this is the current version before use.

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II. The inspection level shall be S-2 and the acceptance number shall be zero.

TABLE II. Sampling for tests.

Component (Unit of product)	Characteristic	Require-ment	Test procedures
Garment fabric*	Weight	3.4	4.4.1
Garment fabric*	Breaking strength, warp	3.5.1	4.4.2
Garment fabric*	Breaking strength, filling	3.5.2	4.4.3
Garment fabric*	Leakage	3.6	4.4.4
Pneumatic trousers	Inflation	3.7	4.4.5
Pneumatic trousers	Pressure, relief valve operation	3.8.1	4.4.6
Pneumatic trousers	Pressure, sustained	3.8.2	4.4.7
Pneumatic trousers	Pressure, proof	3.8.3	4.4.8

* Garment fabric shall be traceable to finished garment material lot.

4.3.1 Classification of defects. Examination shall be conducted in accordance with table III, classification of defects.

Categories	Defects*
Major	
101	Component missing.
102	Component parts not free of sharp edges or corners.
103	Component broken, cracked, torn or cut.
104	Material frayed, scuffed or damaged.
Minor	
201	Case latch does not secure case in closed position.
202	Garment material faded or discolored.
203	Item not free of grease, dirt or foreign material.
204	Case material faded or discolored.

* Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The pneumatic trousers shall be examined for defects in dimensions. Any dimension not within the limits specified herein shall be classified as a defect.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:24Z
Check the source to verify that this is the current version before use.

4.3.3 Packaging inspection. The inspection of the unit package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Weight. The overall weight shall be determined as indicated in Test Method 5040 of Federal Test Method Standard 191.

4.4.2 Breaking strength warp. The garment material shall be tested as specified in Test Method 5102 of Federal Test Method Standard 191.

4.4.3 Breaking strength filling. The garment material shall be tested as specified in Test Method 5102 of Federal Test Method Standard 191.

4.4.4 Leakage. Three specimens (13 inches in diameter) of garment material shall be individually tested on the test jig as shown in figure 2. The specimen shall be placed coated side down, on the holder and the plate collar tightly bolted thereon. Care should be taken to ensure a leaktight fit. Sufficient water shall be poured on top of the specimen to keep the specimen covered at all pressure levels. The specimen shall be subjected to an air pressure of 10 psi for 5 minutes. The air bubbles on the cloth surface produced by air pressure closing the spaces between the cloth and coating shall be removed.

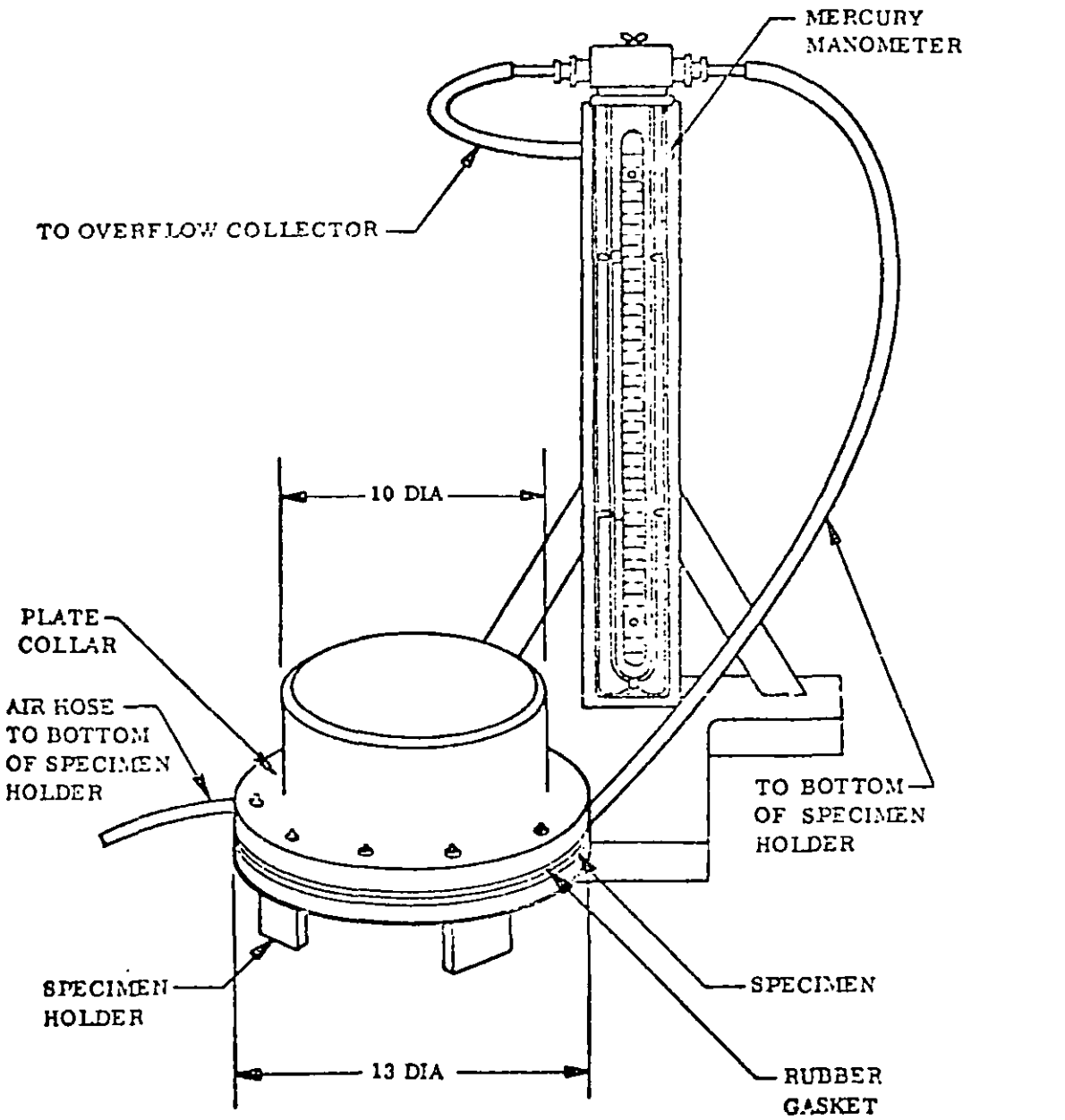
4.4.5 Inflation test. The pneumatic trousers shall be mounted on a "free standing" male manakin and the parts interconnected as described in the vendor operating instructions. The foot pump shall be operated to pressurize the garment to 104 mm of Hg.

4.4.6 Relief valve operation test. The pneumatic trousers shall be mounted and inflated as indicated in 4.4.5 until each relief valve operates.

4.4.7 Pressure, sustained test. The pneumatic trousers shall be mounted and inflated as indicated in 4.4.5.

4.4.8 Pressure, proof test. The outlets of the relief valve shall be taped shut. The pneumatic trousers shall be mounted and inflated as indicated in 4.4.5, except that the trousers shall be inflated to 130 mm of Hg.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:24Z
Check the source to verify that this is the current version before use.



DIMENSIONS IN INCHES

FIGURE 2. Air leakage test jig

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:24Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Unit of issue. One pair of pneumatic trousers, air pump, inflation tubing, Y-connector, on-off valve, instructions and carrying case constitutes one unit of issue.

5.1.2 Level A. Each pair of trousers shall be neatly folded. Tubing shall be coiled in a manner to prevent kinking. The pump, tubing with the Y-connector and on-off valve and folded trousers shall be neatly arranged in the carrying case and the case secured or latched. The case shall be further packaged in a box conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.1.3 Level C. Each pair of trousers along with air pump, inflation tubing Y-connector and on-off valve shall be neatly arranged in the carrying case and secured. The case shall be further packaged in a manner that will afford adequate protection against physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. Four unit packages shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Four unit packages shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged pneumatic trousers shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the carrier's rules and regulations.

5.3 Marking. Each unit package and shipping container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Intended use. The pneumatic trousers covered by this specification is intended for applying pneumatic pressure to the abdomen and legs for use in emergency first-aid treatment to relieve shock, restore blood pressure and counteract internal bleeding.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:24Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) National Stock Number (NSN).

(c) Selection of applicable levels of packing and packaging (see 5.1 and 5.2).

6.3 This specification covers the following item appearing in the ederal Supply Catalog:

National Stock Number	Item Identification
6515-01-023-4531	TROUSERS, PNEUMATIC, ANTI-SHOCK, Universal Size, Blue

6.4 This specification does not include all types, classes, grades or izes of the commodity indicated by the title of this specification, or those hich are commercially available, but is intended to cover the types normally rocured to meet military requirements.

Preparing activity
DLA-DM

Project No. 6515-2094

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:24Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER

MIL-T-37742 (DLA-DM) TROUSERS, PNEUMATIC, ANTI-SHOCK, Universal Size, Blue

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*	TELEPHONE NO.
	DATE

DD FORM 1426 1 OCT 76 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:24Z
Check the source to verify that this is the current version before use.

FOLD

DEFENSE PERSONNEL SUPPORT CENTER
2800 SOUTH 20TH STREET
PHILADELPHIA, PA 19101

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID



HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER
ATTN: DIRECTORATE OF MEDICAL MATERIEL, CODE ATT
2800 SOUTH 20TH STREET
PHILADELPHIA, PA 19101

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:24Z
Check the source to verify that this is the current version before use.